Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks

Jun-Yan Zhu*, Taesung Park*, Phillip Isola, Alexei A. Efros
[1]Berkeley AI Research (BAIR) Laboratory, University of California, Berkeley

Image-to-Image translation is the task of converting an image from one form to another by learning to capture special characteristics of one image collection (domain) and figuring out how these characteristics could be translated into the other image collection. Traditionally, Image-to-Image translation requires paired image data, i.e., an input image being mapped to an output image. For example, a photograph of a scene would be paired with a painting of the same scene, with the goal of translating the photograph into the style of the painting. However, obtaining such paired data can be difficult and costly.

The proposed method of the original paper[10] introduces an approach that learns to translate images from the source domain (X) to the target domain (Y) without paired examples. The method assumes that the two domains share an underlying relationship and learns this mapping through an adversarial loss, ensuring that the translated image is indistinguishable from the target domain.

But as the model is learning to generate images from the source domain to the target one, the problem of mode collapse arises. That mode collapse refers to the situation where all input images are mapped to the same output.[4] This may result in a lack of variety in the generated images, with the model producing very similar or identical outputs regardless of the input. Thus, to ensure meaningful mappings, the paper also introduces an inverse mapping and a cycle consistency loss[9], which ensures that translating an image back to its original domain yields a result close to the original image[1]. The method is applied to various tasks, including style transfer, object transformation, season transfer, and photo enhancement, and it outperforms prior methods in these tasks.

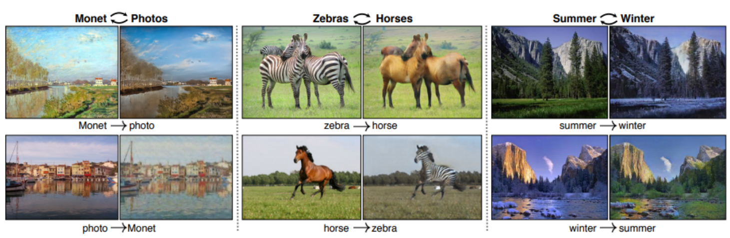

Figure 1: Various examples of Image-to-Image Translation using CycleGAN

The goal is to learn two mapping functions between domains X and Y using unpaired training data. These mappings, $G : X \to Y$ and $F : Y \to X$, are learned using two types of losses: adversarial losses[5] and cycle consistency losses[9].

The adversarial loss[5] is applied to both mapping functions to ensure the generated images resemble those from the target domain. For the mapping $G : X \to Y$ and its discriminator D_Y, the objective is:

$$L_{GAN}(G, D_Y, X, Y) = E_{y \sim p_{\text{data}}(y)}[\log D_Y(y)] + E_{x \sim p_{\text{data}}(x)}[\log(1 - D_Y(G(x)))]$$

Similarly, a similar adversarial loss[5] is applied for the mapping $F : Y \to X$ and its discriminator D_X.

To prevent mode collapse, cycle consistency loss[9]ensures that translating an image to the target domain and back to the original domain results in the original image. The cycle consistency loss[9] is formulated as:

$$L_{\text{cyc}}(G, F) = E_{x \sim p_{\text{data}}(x)}[\|F(G(x)) - x\|_1] + E_{y \sim p_{\text{data}}(y)}[\|G(F(y)) - y\|_1]$$

This loss ensures both forward and backward consistency for the mappings G and F.

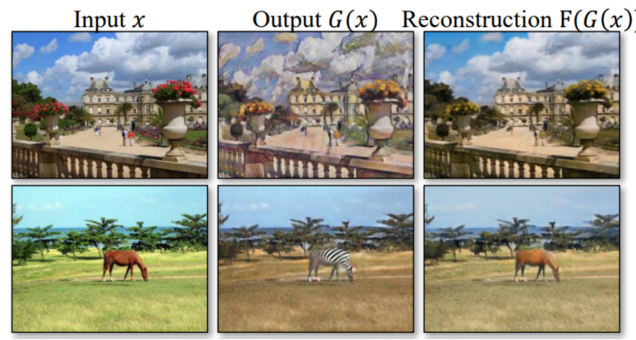

Figure 2: Reconstruction of CycleGAN

The full objective combines the adversarial loss [5]and cycle consistency loss[9]:

$$L(G, F, D_X, D_Y) = L_{GAN}(G, D_Y, X, Y) + L_{GAN}(F, D_X, Y, X) + \lambda L_{\text{cyc}}(G, F)$$

Where λ controls the weight of the cycle consistency loss. Moreover, the implementation details are as below:

- Network Architecture: The architecture is based on Johnson et al.'s method, using residual blocks and PatchGAN discriminators that evaluate overlapping 70x70 image patches.
- Training: The model uses a least-squares loss for stability during training and an image buffer to stabilize discriminator updates. The Adam optimizer is used with a learning rate of 0.0002, decaying after 100 epochs.
- Training Objective: The model is trained to minimize the total loss, optimizing the mappings G and F while simultaneously maximizing the adversarial loss for the discriminators D_X and D_Y.

Upon analyzing the results by the aforementioned implementation, it can be seen that CycleGAN performs remarkably well despite using no paired training data, often producing translations that are close in quality to the fully supervised pix2pix model[6]. When evaluated on paired datasets such as Cityscapes[2] and Google Maps, CycleGAN[10] consistently outperforms other unsupervised baselines, including CoGAN[7], BiGAN/ALI[3], SimGAN[8], and feature-loss-based models. In human perceptual studies, CycleGAN[10] fooled participants in roughly a quarter of the trials, while competing methods almost never fooled them. Quantitative evaluations using the FCN semantic segmentation metric also show that CycleGAN [10] produces more interpretable and realistic images. The full CycleGAN [10]objective is therefore necessary to learn stable and meaningful mappings between domains.

Moreover, although the method works well for tasks involving color or texture changes, it struggles with translations that require geometric transformations, such as changing the shape of objects (e.g., dog → cat). Some failures arise from limitations in the training data; for example, the model becomes confused when encountering images not represented in the dataset, such as people riding horses or zebras. There also remains a performance gap between models trained with paired data and those trained with unpaired data, and fully closing this gap may require introducing weak or semi-supervised signals. Despite these limitations, unpaired data is abundant in many real-world cases, and CycleGAN [10]demonstrates strong progress toward effective image translation without paired supervision.

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